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$$\int_0^1 \frac{\ln x}{\sqrt{x}} dx = \lim_{t \rightarrow 0} \int_t^1 \frac{\ln x}{\sqrt{x}} dx$$

$$\int \frac{\ln x}{\sqrt{x}} dx = \ln x \cdot 2\sqrt{x} - \int \frac{2\sqrt{x}}{x} dx \Rightarrow \left[2\sqrt{x} \ln x - 4\sqrt{x} \right]_t^1 \Rightarrow$$

$$\Rightarrow \lim_{t \rightarrow 0} \left[2\sqrt{x} \ln x - 4\sqrt{x} \right]_t^1 \Rightarrow \lim_{t \rightarrow 0} (-4 - 2\sqrt{t} \ln t + 4\sqrt{t}) \Rightarrow$$

$$\Rightarrow -4 - 2 \cdot 0 \cdot \ln 0 = -4 \quad \text{Integral is convergent}$$

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$$\int_{-2}^3 \frac{1}{x^4} dx = \int_{-2}^0 \frac{1}{x^4} dx + \int_0^3 \frac{1}{x^4} dx = \lim_{t \rightarrow 0} \left(\int_{-2}^t \frac{1}{x^4} dx + \int_t^3 \frac{1}{x^4} dx \right) =$$

$$= \lim_{t \rightarrow 0} \left(\left[-\frac{5}{x^3} \right]_{-2}^t + \left[-\frac{5}{x^3} \right]_t^3 \right) = \lim_{t \rightarrow 0} \left(\left[-\frac{5}{t^3} - \frac{5}{32} \right] + \left[-\frac{5}{3^3} + \frac{5}{t^3} \right] \right) =$$

$$= -\frac{5}{2^3} - \frac{5}{3^3} \approx -0,1468$$

Integral is Convergent

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$$x = \frac{y^4}{8} + \frac{1}{4y^2}$$

$$f(y) = x$$

$$f'(y) = \frac{1}{2}y^3 - \frac{1}{2}y^{-3}$$

$$1 + \left(\frac{1}{2}y^3 - \frac{1}{2}y^{-3} \right)^2 =$$

$$= 1 + \frac{y^6}{4} - \frac{1}{2} + \frac{y^{-6}}{4} =$$

$$= \frac{y^6}{4} + \frac{1}{2} + \frac{y^{-6}}{4} = \left(\frac{1}{2}y^3 + \frac{1}{2}y^{-3} \right)^2$$

$$\int_1^2 \sqrt{\left(\frac{1}{2}y^3 + \frac{1}{2}y^{-3} \right)^2} dy = \left[\frac{y^4}{8} - \frac{y^{-2}}{4} \right]_1^2 =$$

$$= \frac{2^4}{2^3} - \frac{2^{-2}}{2^2} - \frac{1}{8} + \frac{1}{4} =$$

$$= 2 - \frac{1}{16} - \frac{2}{16} + \frac{4}{16} = \frac{33}{16} \approx 2,065$$

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$$y = 1 - x^2$$

$$y' = -2x$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + 4x^2$$

$$S = 2\pi \int_0^1 x \sqrt{1 + 4x^2} dx =$$

$$= \frac{\pi}{4} \int_0^1 8x \sqrt{4x^2 + 1} dx =$$

$$\boxed{\begin{array}{l} 4x^2 + 1 = u \\ du = 8x dx \end{array}}$$

$$= \frac{\pi}{4} \int_0^1 \sqrt{u} du = \frac{\pi}{4} \int_1^5 \sqrt{u} du =$$

$$= \frac{\pi}{4} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^5 = \frac{\pi}{6} \left(5^{\frac{3}{2}} - 1 \right) =$$

$$= \frac{\pi}{6} (5\sqrt{5} - 1)$$